FUELLING INDIA'S TOMORROW

THE EFFORT TO CUT CARBON EMISSIONS BY REDUCING THE VOLUME OF FOSSIL FUELS RELIES ON GREEN HYDROGEN BECOMING A CRUCIAL SOURCE OF CLEAN POWER

The worldwide effort to cut down carbon emissions has acquired urgency in recent years. At the centre of the plan is reducing the use of fossil fuels such as coal, natural gas, petrol and diesel in industry and daily life. Green hydrogen featured in many an emission-reduction pledge at the 2021 UN Climate Change Conference (COP26) in Glasgow, Scotland, the grand plan being to decarbonise heavy industry, freight, shipping and aviation.

Hydrogen has many uses—in fuel cells to generate electricity, in petroleum refining and in fertiliser production. It is also fast emerging as a transportation fuel. Producing hydrogen involves splitting water into its basic components—hydrogen and oxygen—through electrolysis. When this process is conducted using alternative sources of energy such as wind or solar power, the product is called "green hydrogen" as opposed to "grey hydrogen", if the power used to produce it is conventional fuel like coal.

WHY IT IS A GAME CHANGER

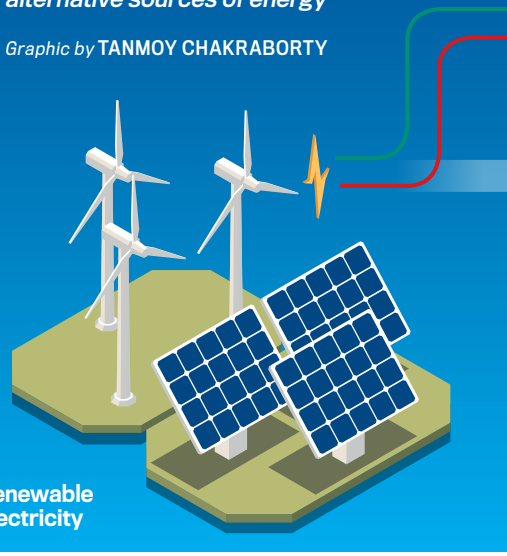
The World Economic Forum describes hydrogen as an important pillar of the net zero economy. The Green Hydrogen Catapult, a UN initiative to lower the cost of green hydrogen, has nearly doubled its goal for electrolysers from 25 gigawatts (GW) set in 2020, to 45 GW by 2027. In January 2023, India approved the National Green Hydrogen Mission with an outlay of Rs 19,744 crore between 2023-24 and 2029-30 to make the country a global hub for the production, use and export of green hydrogen and its derivatives. It has set an ambitious target: green hydrogen production capacity of 5 million tonnes per annum, with an associated renewable energy capacity addition of about 125 GW, by 2030. The goal is to achieve energy independence by 2047 and net-zero emissions by 2070.

Currently, India imports 85 per cent of its crude oil requirement—it bought 232.4 million tonnes in 2022-23, spending

HOW GREEN HYDROG

Producing hydrogen involves splitting water into its basic components–hydrogen and oxygen–through electrolysis, using alternative sources of energy

Graphic by TANMOY CHAKRABORTY



THE CHANGE-MAKERS

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- Plans to commission an integrated green hydrogen ecosystem aimed at generating low-cost renewable power and in turn produce globally competitive green hydrogen and its derivatives at scale
- Its first project of 1 MTPA (million tonnes per annum) is being implemented phase-wise in Gujarat. The aim is to go up to 3 MTPA with about $50 billion (Rs 4.16



"The demand-supply gap for electrolysers used to produce green hydrogen is very wide. We are aiming for a sizeable market share in this. We also want to get into building green hydrogen and green ammonia plants"

–DEREK M. SHAH
Sr Vice President and Head - Green Manufacturing & Development, L&T